

- 2 The photograph shows a garden snail, *Helix aspersa*. This species is endemic to western Europe.



(Source: © Max Mumby / Alamy Stock Photo)

Snails feed on plants in warm, moist conditions.

In dry conditions, snails seal the entrance to their shell with a membrane.

The photograph shows this membrane.



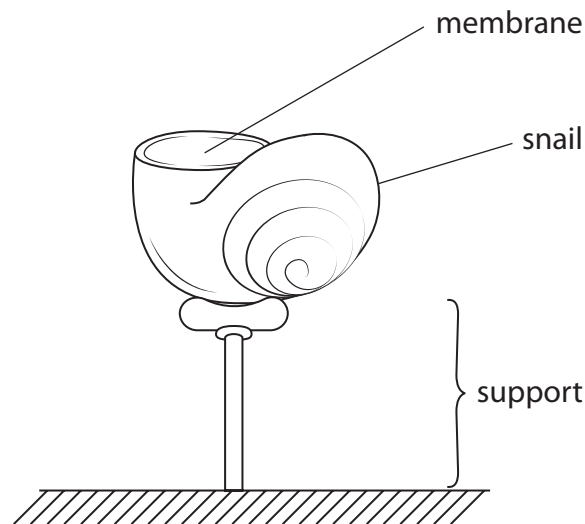
(Source: © Premaphotos / Alamy Stock Photo)

A student investigated the volume of water needed to stimulate snails to break through the membranes at the entrance to their shells.

The student used the following method.

A group of 15 snails with membranes was used in this investigation.

The diagram shows how each snail was held upside down.



Each snail had two drops of water added to the membrane every minute.

The time taken for each snail to break through the membrane was recorded.

- (a) Suggest **one** reason why the student thought it was ethical to use snails in this investigation.

(1)

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(b) The table shows the results of this investigation.

Number of drops of water added	40	80	120	160	200
Total number of snails breaking through the membrane	1	4	7	9	10

The mean diameter of each spherical drop of water was 3.0 mm.

Calculate the volume of water added to a snail in one minute.

The volume of a sphere can be calculated using the formula:

$$V = \frac{4}{3} \pi r^3$$

Give your answer in mm³, to two significant figures.

(3)

Answer mm³

(c) (i) Give **two** abiotic variables, other than the independent variable, that could affect this experiment.

(2)

1

2

(ii) Name **one** of the variables you have identified in (i).

Variable

State how this variable could be controlled.

(1)

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(iii) Suggest the effect that not controlling this variable could have on the results.

(1)

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- (d) The student decided to repeat the investigation using the same apparatus but adding four drops of water every minute.

There was no significant difference in the results when the number of drops added each minute doubled.

Suggest why this did **not** have an effect on the time it took the snails to break through the membrane.

(2)

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(Total for Question 2 = 10 marks)